



# CYCRAI 2027

## FIRST INTERNATIONAL CONFERENCE ON CYBER SECURITY, CRYPTOGRAPHY AND RESPONSIBLE AI



Cyber  
Security



Cryptography



Responsible  
AI



29 - 30 SEPTEMBER 2027



SCAN THE QR CODE  
FOR MORE INFORMATION

## ABOUT CYCRAI

First International Conference on Cyber Security, Cryptography and Responsible AI (CYCRAI 2027) will be organized by Department of Computer Science and Engineering (Cyber Security), Institute of Engineering & Management Kolkata, under University of Engineering and Management Kolkata, with Technical Co-Sponsorship of IEEE Kolkata section, IEEE Computer Society SBC UEM Kolkata and IEEE Communication Society Kolkata Section during 29th September to 30th September, 2027.

\*(All accepted and presented papers will be submitted for inclusion in the **IEEE Xplore Digital Library** subject to meeting IEEE Xplore's scope and quality requirements-Approval Awaiting)\*

## ABOUT THE DEPARTMENT

The Department of Computer Science and Engineering (Cyber Security) programme at IEM Kolkata is allowing students to gain knowledge about concepts that enable many of today's fundamental technologies, including artificial intelligence & machine learning, Wireless Communication and the Internet of Things. The Department introduces fundamental concepts of Computer Science, Information Technology, Cybersecurity and Networks and imparts knowledge dealing with logical reasoning, problem-solving, data representation, abstraction etc. The initial level courses provide students with a core foundation in programming, mathematical reasoning, physics, and circuit design. The later part blends fundamentals in computer technologies including software engineering, database management system, computer networks, operating systems, and computer architecture with fundamentals in electrical and electronics engineering. Currently, the Department runs a four-year B. Tech in Computer Science and Engineering (Cyber Security) course to prepare students with a firm foundation of both the theory and practice of Computer Science and Engineering into program and to strengthen creativity, nurture innovation and develop the ability to carry out research and solve real-world problems.

## PAPER SUBMISSION LINK

All submissions to CYCRAI 2027 must be made through Microsoft CMT.

SubmissionPortal:

<https://cmt3.research.microsoft.com/CYCRAI2027>

- Authors are requested to create an account on Microsoft CMT before making a submission
- Submit your paper, track its status, and respond to reviews through the CMT dashboard.
- All communications regarding submission will be handled exclusively via Microsoft CMT.
- For any queries related to submission, please use the contact information provided on the conference website

### IMPORTANT DATES

| EVENT                          | DATE                   |
|--------------------------------|------------------------|
| Full Paper Submission Starts   | 15 December 2026       |
| Full Paper Submission Deadline | 15 March 2027          |
| Acceptance Notification        | 15 June 2027           |
| Author Registration Deadline   | 10 August 2027         |
| Camera Ready Submission        | 28 August 2027         |
| Conference Dates               | 29 – 30 September 2027 |

## TRACK STRUCTURE



**TRACK 1**  
FOUNDATION OF  
CYBER SECURITY

Track Chair:  
Prof. Shreya Nag



**TRACK 2**  
FOUNDATION OF  
CRYPTOGRAPHY

Track Chair:  
Prof. Siddhartha Roy



**TRACK 3**  
FOUNDATION OF  
RESPONSIBLE AI

Track Chair:  
Prof. Arnab Ghosh

Detailed topics under each track are available on the website.

### TRACK 1 – FOUNDATION OF CYBER SECURITY

1. Cybersecurity Awareness
2. Cybersecurity Governance
3. Risk and Compliance
4. Digital Forensics and Cybercrime Investigation
5. Cyberattack Reconstruction and Attribution
6. Hardware Security and Reliability
7. IoT and Edge Security
8. Identity, Ethics and
9. Databases in Cyber and Privacy
10. Malware Analysis
11. Threat Modelling, Security and Privacy Preserving ML
12. Digital Twin Security
13. OS & Firmware Security for Recovery
14. Secure Messages and CE
15. Cybercrime Investigation and Prevention
16. Vulnerability Scanning and Prevention
17. Zero Trust Architecture and Prevention
18. Cyber-Physical Distributed Ledger Security
19. Security and Defense
20. AI-Driven Cyber Defence Development

### TRACK 2 – FOUNDATION OF CRYPTOGRAPHY

1. Cryptography and Cryptanalysis
2. Information Theory and Cryptography
3. Applied Cryptography and Network Security
4. AI/ML for Cryptography
5. Lightweight Cryptography for IoT and Embedded Systems
6. Applied Cryptography and Its Implementation
7. Quantum Cryptography
8. Quantum Machine Learning
9. Quantum Computing
10. Post Quantum Cryptography

### TRACK 3 – FOUNDATION OF RESPONSIBLE AI

1. AI Ethics
2. AI Fairness
3. AI Bias Mitigation
4. Explainable AI
5. Interpretability in AI Audit
6. Privacy-Preserving AI Models
7. Secure AI Systems
8. Responsible Generative AI
9. AI for Risk-Informed Analytics
10. AI Security in Healthcare, Education, Industry, and Smart City Framework
11. Ethics in Trust and Human-AI
12. Societal Impact of Responsible AI Development
13. Security and Privacy for AI Systems
14. AML for Threat Detection and Prevention

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